Baltic Energy Transit Risk

Exposure of post-2022 energy import infrastructure

Tobyn Smith

Executive summary

The Baltic region replaced a continental pipeline network with five working coastal sites. Russian pipeline gas into the eight littoral states fell from roughly 70 bcm in 2021 to 0.3 bcm in 2023, while the terminals built to receive seaborne gas hold about 23.4 bcm a year, some 33% of what stopped arriving.

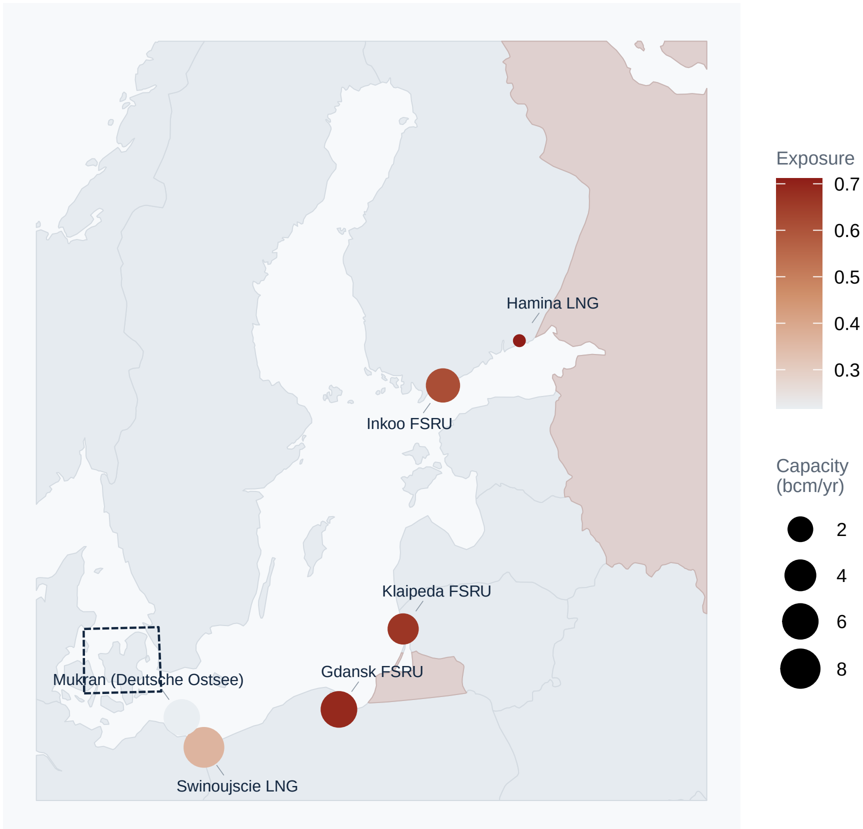
The strategic problem is not the size of that gap, which lower demand and pipeline gas from the west and north have largely closed. It is that the infrastructure which replaced the pipelines is concentrated, sits nearest Russian territory precisely where it is smallest, and is one asset deep in every country that holds it. Policymakers should therefore treat these terminals less as a solved substitution and more as a set of single points of failure whose loss would push the region back onto the same subsea links that have already been cut three times since 2022.

This brief sets out where that infrastructure sits, how exposed each terminal is on a stated and adjustable measure, what happens when the largest terminal in each country is removed, and what follows for policy.

Where the terminals sit

Baltic LNG import terminals

Point size is capacity, colour is exposure; dashed box is the Danish Straits



Sources: Natural Earth; operator reporting and Global Energy Monitor.

Findings

Capacity is concentrated in the south. Świnoujście in Poland and Mukran in Germany hold about three fifths of the region's operating import capacity. The northeast leans on one or two terminals each, Inkoo for the Gulf of Finland and Klaipėda for Lithuania. What replaced a continental pipeline network is, in practice, five working coastal sites, with a sixth planned at Gdańsk.

The terminals nearest Russia are the small ones. Hamina, at the eastern end of the Gulf of Finland, and Klaipėda and Gdańsk beside the Kaliningrad exclave sit closest. The two largest terminals are among the farthest away. Size and proximity do not line up, which is precisely what the exposure score is built to expose.

Every cargo enters through one strait. All LNG bound for these terminals passes the Danish Straits. The Kiel Canal is the only other way in and is far too small for LNG carriers. The terminals deepest inside the sea run the longest exposed route to port.

The small states overbuilt on purpose. Set against what each country actually burns, Finland and Lithuania hold capacity worth several times their own demand, while Germany and Poland cover only part of theirs by sea. Across the eight countries the terminals could land 22% of annual consumption.

Table 1: Terminals ranked by exposure score.

Terminal	Country	Status	Capacity (bcm)	km to Russia	Exposure
Hamina LNG	Finland	operating	0.30	34	0.71

Terminal	Country	Status	Capacity (bcm)	km to Russia	Exposure
Gdansk FSRU	Poland	planned	6.10	62	0.69
Klaipeda FSRU	Lithuania	operating	3.75	50	0.67
Inkoo FSRU	Finland	operating	5.00	216	0.61
Swinoujscie LNG	Poland	operating	8.30	353	0.37
Mukran (Deutsche Ostsee)	Germany	operating	6.00	388	0.22

What follows for policy

1. **The northeast has no spare door.** Borrowing the logic of the EU's N-1 standard and removing each country's largest terminal, Finland is left covering about a fifth of its demand and Lithuania, Poland and Germany are left with no seaborne capacity at all. They are not failing the real test, which counts pipelines and storage too, but the fallback is exactly the links that have been damaged since 2022.
2. **The connectors matter as much as the terminals.** Estonia, Latvia and Sweden import through neighbours rather than terminals of their own, over links such as the Balticconnector that have already been damaged.
3. **The Danish Straits are a shared dependency.** One entrance serves all eight countries, so surveillance and patrol arrangements there protect the whole region at once.
4. **Floating terminals cut both ways.** They were quick to add and can be moved, but they are few and chartered. Mukran lost half its capacity when a charter ended.

This is not hypothetical

Date	Asset	What happened
Sep 2022	Nord Stream 1 and 2	Pipelines ruptured by underwater explosions off Bornholm
Oct 2023	Balticconnector	Pipeline and cable damaged by a ship's anchor; back in service 22 April 2024
Dec 2024	Estlink 2	Power and telecoms cables cut; back in service 20 June 2025

Limitations

This measures geographic exposure, not the probability of disruption. The score uses proxies rather than a causal model, and different reasonable weights shift the ranking. It is a single snapshot: capacities move, and headline figures often describe planned rather than operating capacity. Demand in 2023 was itself depressed by high prices and mild weather, so coverage in a colder year would look less comfortable.

Sources

All public. Terminal locations and capacities from operator reporting and the Global Energy Monitor LNG tracker, current to 2024 and 2025. Gas import dependency (`nrg_ind_id`), inland consumption (`nrg_cb_gas`) and imports from Russia (`nrg_ti_gas`) from Eurostat. Country geography from Natural Earth.

Full data, methods and interactive maps: https://tobyn-smith.github.io/transit